



F1-19 Flow Channel

The Flow Channel introduces students to the characteristics of flow in an open channel at an elementary level.

- Demonstrating basic phenomena associated with open channel flow
- Visualisation of flow patterns over or around an immersed object

PRODUCT CODE: F1-19

Categories: Agricultural, Educational, F1 Series Fluid Mechanics

Tags: Armfield , Asymmetrical Aerofoil , Broad Crested Weirs , Dye Injection , F1-19 , Flow Channel , Flow Patterns , Fluid Mechanics , Hydraulics Bench , Open Channel Flow , Overshot Weir , Sharp Crested Weir , Symmetrical Aerofoil , Undershot Weir , Visualisation Of Flow Patterns

DESCRIPTION

The channel consists of a clear acrylic working section of large depth-to-width ratio incorporating undershot and overshot weirs at the inlet and discharge ends respectively. Water is fed to the streamlined channel entry via a stilling tank to reduce turbulence. Water discharging from the channel is collected in the volumetric tank of the hydraulics bench and returned to the sump for recirculation.

A dye injection system incorporated at the inlet to the channel enables flow visualisation in conjunction with a graticule on the rear face of the channel.

Models supplied with the channel include broad and sharp-crested weirs, large and small-diameter cylinders and symmetrical and asymmetrical aerofoils. These in conjunction with the inlet and discharge weirs, permit a varied range of open channel and flow visualisation demonstrations, Adjustable feet permit levelling.

TECHNICAL SPECIFICATIONS

Diameter of test pipe: 3.0mm

Length of test pipe: 760mm

Distance between pressure tapping points: 500mm

Range of mercury manometer: 500mm

Range of water manometer: 500mm

Measuring cylinder capacity: 1000ml



FEATURES & BENEFITS

Experimental content

- To visualise flow patterns around immersed objects in an open channel
- Demonstration of flow phenomena in an open channel
- Undershot and Overshot weirs
- Broad Crested and Sharp Crested Weirs
- Discharge beneath a sluice gate
- Creation of a hydraulic jump downstream of a sluice gate and weir
- Drowning of a sluice gate and weir
- Flow over a broad crested and sharp edged weir
- Supercritical (fast) and sub-critical (slow) flows over the weir

Video (English & Spanish)

Watch the F1-19 in action:



Requirements

F1-10 Hydraulics Bench to provide water flow.

Dimensions

Length: 0.865m

Width: 0.33m

Height: 0.50m

Order Codes

F1-19

YOU MAY ALSO LIKE...



**F1-18-MKII
ENERGY LOSSES
IN PIPES**

[Read more](#)



**F1-10
HYDRAULICS
BENCH**

[Read more](#)



**F1-23-MKII
FREE AND
FORCED
VORTICES**

[Read more](#)



**F1-12-MKII
HYDROSTATIC
PRESSURE**

[Read more](#)



Armfield Limited. Unit 10 Headlands Business Park, Salisbury Road, Ringwood, Hampshire BH24 3PB.
England. Tel: +44(0) 1425 478781

